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Magnetized white dwarfs with mixed poloidal-toroidal fields

Draft – structure and core physics only

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ABSTRACT

Context. TBD.
Aims. TBD.
Methods. TBD.
Results. TBD.

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1. Introduction

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2. Equation of state

The background stellar structure throughout this work is that of a fully degenerate, zero-temperature electron gas (ztwd; Chandrasekhar 1935). Pressure and density are both parametrized by the normalized Fermi momentum $x \equiv p_F/(m_e c)$,

$$P(x) = A\left[x(2x^2-3)\sqrt{x^2+1} + 3\sinh^{-1} x\right], \qquad (1)$$

$$\rho(x) = B\,x^3, \qquad (2)$$

where A is a fixed combination of fundamental constants (m_e, c, $\hbar$) and B depends on the composition only through the mean molecular weight per electron $\mu_e = 1/Y_e$, with Y_e the number of electrons per baryon. This is a purely mechanical equation of state: at fixed composition, pressure depends on density alone, with no temperature dependence. Consequently, once the configuration is restricted to be non-rotating and field-free, the entire stellar structure (mass, radius, density profile) is determined by exactly two physical parameters: the central density ρ_c and μ_e.

The specific enthalpy has a closed form,

$$H(x) = \frac{8A}{B}\left[\sqrt{1+x^2} - 1\right], \qquad (3)$$

which is what makes the self-consistent field (SCF) iteration of Sect. 3 tractable: at each iteration the density is recovered directly from the local enthalpy via the inverse relation $x(H) = \sqrt{(1+HB/8A)^2 - 1}$, with $H \leq 0$ mapped to $\rho = 0$ (the stellar surface).

Near the surface ($x \ll 1$) the equation of state reduces to $\rho \propto H^{3/2}$, an effective polytropic index $n = 3/2$; in the relativistic core ($x \gg 1$) it approaches $\rho \propto H^3$, $n = 3$ – the marginal polytropic index at which the equilibrium mass becomes independent of central density, the origin of the Chandrasekhar mass limit.

At sufficiently high density, inverse beta decay (electron capture) changes the composition locally and the constant-μ_e cold equation of state above ceases to describe a real white dwarf. We adopt the tabulated threshold densities of Boshkayev et al. (2013) for this validity limit; all central densities discussed below are kept below this threshold for the assumed composition.

3. Field equations

Axisymmetric magnetohydrostatic equilibria with a poloidal field are governed by the Grad-Shafranov equation (Grad & Rubin 1958; Shafranov 1966) for the poloidal flux function $u(\varpi, z) = \varpi A_\varphi$, where ϖ is the cylindrical radius and A_φ the azimuthal component of the vector potential. The poloidal field components follow from u as

$$B_\varpi = -\frac{1}{\varpi}\frac{\partial u}{\partial z}, \qquad B_z = \frac{1}{\varpi}\frac{\partial u}{\partial \varpi}, \qquad (4)$$

and the equilibrium (force balance combined with the Poisson equation for self-gravity) reduces to a single elliptic equation for u sourced by the current distribution and, self-consistently, by u itself through the poloidal current function $M(u)$.

Following the SCF approach of Hachisu (1986), extended to magnetized configurations, the poloidal current function is taken as linear in the flux, $M(u) = k_0\,u$, so that the Bernoulli integral governing hydrostatic balance reads

$$H + \Phi - M(u) = C, \qquad (5)$$

with Φ the gravitational potential and C a constant fixed by the boundary condition at the stellar surface. Equation 5 and the Grad-Shafranov equation are solved iteratively together with the Poisson equation for Φ, alternating between an update of the density field (from the equation of state, Sect. 2) and an update of the flux function u, until convergence.

A toroidal field confined within the last closed poloidal field line ($u > u_c$) is added through a toroidal current function $\beta(u)$, parametrized as $\beta(u) = \zeta(u - u_c)^{m+1}$ with $m \geq 1$, giving $B_\varphi = \beta(u)/\varpi$ inside the closed-field region and zero outside. A purely toroidal, non-mixed branch is also considered separately, with $B_\varphi = K\rho^m\varpi^{2m-1}$ derived directly from the Lorentz force (Maxwell stress tensor) rather than from a flux function; this branch requires no Grad-Shafranov solve, at the cost of being unable to represent a poloidal component. A genuinely self-consistent *mixed* poloidal-toroidal equilibrium – both components simultaneously present and mutually consistent within the same SCF solve – is outside the scope of the axisymmetric formulation used here; Sect. 5 describes how mixed configurations are instead obtained dynamically.

Purely toroidal and purely poloidal equilibria are each subject to a non-axisymmetric instability of azimuthal wavenumber $m = 1$ (Tayler 1973; Markey & Tayler 1973), which the axisymmetric formulation of this section cannot, by construction, test for.

4. Castro

Dynamical (as opposed to SCF-equilibrium) evolution is carried out with Castro (Almgren et al. 2010), a massively parallel, block-structured adaptive-mesh-refinement (AMR) code for compressible astrophysical flows, built on the AMReX framework (Zhang et al. 2019). The hydrodynamics solver uses an unsplit corner-transport-upwind scheme (Colella 1990) with piecewise-parabolic reconstruction (Colella & Woodward 1984); self-gravity is included for isolated (non-periodic) configurations, and a constrained-transport magnetohydrodynamics (MHD) solver extends the same integrator to evolve a magnetic field while preserving $\nabla\cdot\mathbf{B} = 0$ to machine precision by construction. Castro accepts an arbitrary equation of state and reaction network through the StarKiller Microphysics interface; for this work the network is reduced to a single inert species reproducing the assumed μ_e, and the equation of state is the same zero-temperature degenerate electron gas of Sect. 2, so that a star built in hydrostatic equilibrium by the SCF solver is evolved under the identical thermodynamic relation it was constructed with.

Self-gravity for an isolated spherical mass distribution is obtained from Castro's monopole gravity solver, which integrates the enclosed mass along radial shells. Initial conditions are supplied to Castro as a one-dimensional radial density (and composition) profile, sampled onto the three-dimensional Cartesian AMR grid; the sampling itself (cell-volume averaging versus point sampling, and the alignment of the grid relative to the stellar center) has a measurable effect on how closely the discretized initial condition reproduces the target central density, which we account for when initializing a background star for dynamical evolution.

5. The Braithwaite method

As noted in Sect. 3, the axisymmetric SCF formulation cannot produce a genuinely self-consistent mixed poloidal-toroidal equilibrium, nor can it test the stability of any equilibrium against non-axisymmetric perturbations – a serious limitation given that both purely poloidal and purely toroidal fields are themselves non-axisymmetrically unstable (Tayler 1973; Markey & Tayler 1973). The resolution adopted here follows Braithwaite & Spruit (2004) and Braithwaite & Nordlund (2006): rather than constructing a mixed field and separately testing its stability, a *random*, multi-scale magnetic field is seeded within an otherwise field-free, non-rotating background star (Sect. 2) and allowed to relax dynamically under full three-dimensional, ideal MHD in Castro (Sect. 4). Because the evolution is a genuine dynamical calculation and not an imposed equilibrium, any configuration that survives the relaxation is, by construction, stable to the perturbations sampled by the initial random field – including non-axisymmetric ones the Grad-Shafranov formulation cannot represent at all.

The random seed field is constructed as a vector potential $\mathbf{A}$ – a superposition of Fourier-like modes spanning a band of length scales, with random amplitude, phase, and direction for each mode – multiplied by a smooth radial envelope that confines the field within the stellar volume before the magnetic field itself is formed as the discrete curl, $\mathbf{B} = \nabla \times \mathbf{A}$. Windowing the potential rather than the field is required for $\nabla\cdot\mathbf{B} = 0$ to be preserved exactly: multiplying an already-divergence-free $\mathbf{B}$ by a confinement window after the curl has been taken would reintroduce a spurious divergence, whereas curling an already-windowed potential does not. The total seed field energy is calibrated, prior to relaxation, to a chosen fraction of the star's gravitational binding energy, $E_{\mathrm{mag}}/|W|$.

Once released, the field is left free to evolve under the full MHD equations; no particular final poloidal-to-toroidal ratio is imposed. The relaxed configuration's toroidal fraction, $E_{\mathrm{tor}}/E_{\mathrm{mag}}$, is measured directly from the evolved field, and its dependence on the initial random seed (reproducibility across independent realizations of the random field, at fixed $E_{\mathrm{mag}}/|W|$) is examined empirically rather than assumed.

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